MSc Research Project/Internship/ Practicum

Abstract

Abstract

- **Brief background** (Say what problems are addressing your research question)
- **Why did you do what you did? - why did it need to be done in the field?** (Why do people care about this problem – why this is important)
- **What did you do?** (What is your contribution - highlight your innovation)
- **What happened when you did it?** (Main conclusion of results)
- **What do the results mean in theory** (Describe how your work fits with the current state of the art)
- **What do the results mean in practice** (What is the key benefit)?
- **What remains unresolved?**

Abstract Example (Data Analytics)

Current diagnostic systems to identify cardiovascular diseases (CVDs) such as Echocardiography (ECG) require both time and highly skilled physicians to evaluate complex combinations of clinical and pathological data. - <u>BACKGROUND</u>

Inaccurate decision making is the challenge in the process and thus can't be permitted in healthcare industry. -<u>PROBLEM/MOTIVATION (WHY)</u>

Data mining methodologies can be applied to large medical datasets to extract insights that aid healthcare professionals in the diagnosis of cardiovascular diseases. In data mining, feature selection and classification algorithms are considered as a concern of global "combinatorial optimization". <u>WHAT CAN BE DONE TO ADDRESS THE PROBLEM</u>

This paper investigates the optimal hybrid model of feature selection and classification algorithms in the diagnosis of cardiovascular diseases based on three performance metrics namely accuracy, sensitivity and specificity. –<u>HOW DID YOU DO IT</u>

The word 'optimal' refers to the best or most favorable hybrid models actualized in this paper.

This investigation will thus empower physicians to diagnose cardiovascular diseases and initiate timely treatment without the intervention of a trained cardiologist. -<u>WHAT THE RESULTS MEAN IN PRACTICE</u>

Translated to structured abstract….

Background: Current diagnostic systems to identify cardiovascular diseases (CVDs) such as Echocardiography (ECG) require both time and highly skilled physicians to evaluate complex combinations of clinical and pathological data.

Objectives: The aim of this work is to develop data mining methodologies that can be applied to large medical datasets to extract insights that aid less skilled healthcare professionals with the timely diagnosis of cardiovascular diseases. A hybrid model of feature selection and classification algorithms will be developed to select pertinent features relevant to CVD diagnose and to classify ECGs into risk categories for CVD. Performance of the model will be measured using accuracy, sensitivity and specificity. The word 'optimal' refers to the best or most favorable hybrid models actualized in this paper.

Results: The hybrid model selected 15 features related to CVD from a dataset of 35 and achieved classification accuracy of 71% (sensitivity 90%, specificity 65%).

Findings: In terms of accuracy, model compared well with physicians (71% vs 69%) and with previous work in the field, (Bloggs et al. achieved accuracy of 65%). However the model is overcautious (low specificity) and future work should address this while maintaining or improving sensitivity.

An Abstract from last year…

Background:

Online marketplaces are helping to drive e-commerce sales. Competitive sellers go to great lengths to ensure that their products are noticed. This results in sellers posting the same advertisement several times, using near-duplicate titles or using slightly altered descriptions.

Objectives:

This study proposes to build a dichotomous classifier that would spot duplicate commercial advertisements that feature the same product in an online marketplace.

Methodology:

A Kaggle dataset of 3 million records used in the research. The dataset was imbalanced; data samples for duplicate class was less as compared to the non-duplicate class. Six oversampling technique were employed to achieve class balance in the dataset - Random oversampling, SMOTE, SMOTE-Borderline 1, SMOTE-Borderline 2, SVM SMOTE and ADASYN. Four classification models - Gradient Boosting Tree, Logistic Regression, Naive Bayes and SVM, were built on the sampled data to identify the duplicate advertisements and their results compared.

Results:

The performance of classifiers was found to improve with an increase in the sample size of training data. The best performing model was SVM when paired with Borderline-SMOTE 2, with an F1 score of 0.9151

Findings:

The approach is successful in detecting a high number of duplicates and compared well with previous work in the field (F1 score of 0.89 was achieved by X).